

Person 3 | Dataset: 99 papers, 1,958 chunks

What Was Done

A hybrid search system (built by Person 2) was used to find relevant chunks from 99 AI research papers by combining keyword search (TF-IDF) and semantic search (dense embeddings). The problem was its settings were just guessed — it was unknown if $\alpha=0.7$ and $k=5$ were actually the best values. The goal was to find the best settings automatically using **Optuna**.

Why Optuna?

Instead of manually trying every combination (which would take forever), Optuna tries different settings, learns from each attempt, and gets smarter over time. **40 trials** were run and each time retrieval quality was measured using NDCG@5. This is called **AutoML** — letting the machine tune itself.

Baseline vs AutoML Results

Method	Recall@5	NDCG@5	P95 Latency
Baseline (fixed settings)	0.9000	0.8274	0.0852s
AutoML (Optuna best)	0.9667	0.8479	0.0263s

Better results were found on every metric — and 3x faster.

What Parameters Mattered Most

Parameter	Importance
k (chunks retrieved)	0.521 — most important
svd_dim (compression)	0.214
alpha (dense vs TF-IDF)	0.116
metric	0.088
normalization	0.061

k was by far the most important. Retrieving 15 chunks instead of 5 gave the system more chances to find the right one.

Why Did Results Improve?

- **k=15 instead of 5** — more candidates = better recall
- **alpha=0.644 instead of 0.7** — slightly more TF-IDF weight was found to work better for AI terminology
- **svd_dim=500** — keeping more dimensions was found to preserve retrieval quality
- **normalization=False** — raw vectors were found to keep useful score information that normalization was throwing away

Conclusion

A configuration that beats the baseline on all metrics was found by Optuna. The key lesson: **how many chunks are retrieved (k) matters more than anything else**. The best config was saved in `best_config.yaml`.

Chat Link:

<https://claude.ai/share/299d98f0-dab5-4667-a56c-e84db97e6dc1>